

PROFILE SUMMARY

Master's student in Autonomous Systems at Hochschule Bonn-Rhein-Sieg with a strong background in autonomous mobile robotics, sensor fusion, and AI based robotic perception. Has over three years of experience in developing navigation, perception, and human robot interaction modules for robotic systems. Adept at Python 3 and ROS 2 with real world experience integrating algorithms for autonomous systems. Looking forward to putting myself into more challenges in the world of autonomy, intelligence, and mobility.

TECHNICAL SKILLS

- **Programming:** Python 3, C++, Linux
- **Robotics & Navigation:** ROS 1, ROS 2, SLAM, Path Planning, AMCL, MoveIt
- **Simulators:** Gazebo, Isaac Sim, CARLA, Webots
- **Perception & Vision:** YOLO, OpenCV, LiDAR Processing, PCL
- **Filters:** Extended Kalman Filter, Kalman Filter, Particle Filter
- **Kinematics:** Forward & Inverse Kinematics, DH Parameters
- **Tools & Other:** GitHub, Docker, Fusion 360

PROFESSIONAL EXPERIENCE

iHub Robotics

Robotics Engineer

Cochin, India

Dec 2023–Nov 2024

- Developed and implemented navigation and path planning modules for semi-humanoid and service robots operating in dynamic, unstructured environments.
- Built mapping and localization pipelines by fusing LiDAR and depth-sensor data, and validated the full navigation stack in Gazebo and Isaac Sim before hardware deployment.
- Processed LiDAR and depth point-cloud data using PCL for clustering, obstacle detection, and real-time perception in autonomous robotic systems.
- Integrated face recognition, gesture-based interaction, and voice recognition systems using AI perception frameworks such as OpenCV, PyTorch, and TensorFlow for natural human-robot interaction.
- Deployed real-time perception, sensor fusion, and onboard AI inference on NVIDIA Orin Nano and Jetson platforms for resource-constrained robotic applications.

Bharat Electronics Limited

Graduate Apprenticeship Trainee

Bangalore, India

Dec 2022–May 2023

- Worked in the Network Centric Systems department on quality assurance, testing, and validation of communication and electronic systems.
- Performed functional testing and diagnostics using electronic test equipment and assisted in hardware validation tasks.
- Documented test procedures and results, and collaborated with senior engineers to troubleshoot faults and ensure compliance with quality standards.

SOFTWARE PROJECTS

- **Explainable Anomaly Detection in Assembly Line (Deep Learning)**
Developed a deep learning-based anomaly detection model on the MVTec AD dataset to detect and localize defects in assembly line components, using Grad-CAM for explainable, visual model interpretability suited for industrial quality inspection.
- **Autonomous Racing Car Perception & Planning (BRS Motorsport, HBRS Bonn, ongoing)**
Developing the perception and planning pipeline for an autonomous racing car with the BRS Motorsport Formula Student team, applying ML-based clustering on LiDAR point clouds for cone detection and generating optimal trajectories between cones for autonomous navigation.
- **Human Detection and Task Sequencing (Toyota HSR)**
Built a ROS 2 perception pipeline on the Toyota HSR using depth camera and LiDAR for real-time person and hand-gesture detection, with a custom YOLO model and priority-queue-based task sequencing. Integrated autonomous navigation and manipulator control for goal-directed delivery in an unmapped environment.
- **Custom ROS 2 Navigation Stack**
A ROS 2 based autonomous navigation pipeline integrating LiDAR and depth sensor perception, A* global path planning, local obstacle avoidance, and autonomous exploration when no pre-mapped environment is available.
- **ROSbridge Docker Container**
Developed a Docker-based solution enabling ROS 1-ROS 2 communication via ROSbridge, improving multi-robot integration workflows and data exchange.
- **Semi-Humanoid Robot with AI perception modules**
Developed real-time face recognition and gesture based interaction. Integrated AI inference using NVIDIA Orin Nano for human-robot interaction and behavioral responses.
- **Autonomous GPS Navigation System (Surface Vehicle)**
Created a GPS based autonomous navigation system for waypoint tracking and path following on a boat using GPS and compass data.

RESEARCH

- **Towards Embodiment-Agnostic Human-to-Robot Motion Retargeting** (M.Sc. Autonomous Systems, HBRS)
Authored a position paper surveying human-to-robot motion retargeting for learning from demonstration across geometric, optimisation-based, learning-based, and physics-aware imitation methods. Identified physics feasibility, morphology dependence, and input-quality sensitivity as core open problems, and proposed embodiment-agnostic pipelines that map human motion onto arbitrary robot morphologies without per-robot tuning.

EDUCATION

Hochschule Bonn-Rhein-Sieg
M.Sc. Autonomous Systems

Sankt Augustin, Germany
Mar 2025–Current

APJ Abdul Kalam Technological University
B.Tech. Mechatronics Engineering

Thrissur, India
Aug 2018–Jun 2022

AWARDS

- **Junction Hackathon 2025** — First Prize Winner in Espoo, Finland. Built an intelligent EV charging orchestration platform using GenAI (Gemini) for a DENSO challenge on digital identities for vehicles and charging stations. 2025

LANGUAGES

- **Malayalam:** Native
- **English:** Fluent (C1)
- **German:** A1 (actively learning)